X-SIG Poster Presentation
Summer Research 2023

Highly Efficient Songbird Surveys using Drones

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Introduction

- Birds are commonly used as bioindicators.
- Automated recording units (ARUs) are a popular tool used in bioacoustic bird surveys (recording bird sounds).
- **BUT**, ARUs are time-consuming to deploy.
- Short recordings from drones (Uncrewed Aerial Systems, or UAS) are a proposed alternative ~ airborne bioacoustics.
- **Our Hypothesis: 1-minute airborne bioacoustic surveys will provide comparable data to 10-minute ARU surveys.**

Methods

- Study site: State Game Lands 249 near Biglerville, Adams County, PA.
- Sample design: surveys points on a 200-m grid (Fig. 1)

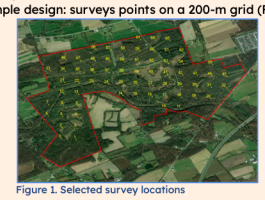


Figure 1. Selected survey locations

- Each point was surveyed twice using ARUs and twice using UASs, 6:30 to 9am, June 2023.
- **ARU Protocol:**
  - on 80cm stake above ground (Fig. 2).
  - 10 mins of audio analyzed.
- **UAS Protocol:**
  - Used Litchi Hub to create waypoints.
  - Drone (Fig 3.) hovered at an altitude of 55m above ground-level.
  - 18m (60ft) fishing line suspended a
  - recorder beneath the drone.
  - Drone hovered at each point for 70 seconds, first 10 seconds not used.
- **Audio Analysis:**
  - Trimmed in Audacity.
  - Uploaded and analyzed using AI Merlin.
  - Data manually verified by Dr. Andrew Wilson.
- **Statistical Analysis:**
  - It is known that not every bird present is detected.
  - Occupancy modeling to estimate proportion of points occupied by each species.
  - Analysis in R using "unmarked" package.



Figure 2. Songmeter ARU in the field



Figure 3. DJI Mavic

Results

- 71 species were detected in all, with slightly more on ARUs than UAS (Table 1).
- More species were detected on 1-minute UAS surveys than in the first minute of ARU surveys (Table 1).

	ARU		UAS
	10-min	1-min	1-min
Species detected	62	47	59
% of all species	87.3	66.2	83.1

Table 1. Total bird species detected across of survey points by the two methods

- Most species were detected at more points by ARUs than UAS (Fig. 4), but this was likely due to the longer recording period.
- For 72% of species, ARU detection was higher than UAS.
- **BUT,** UAS had more species detections than the first minutes of ARUs 59% of species (Fig 5.)

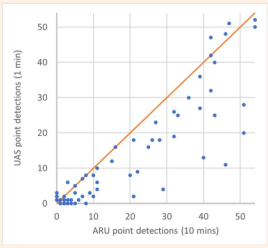


Figure 4. Number of points at which each species (n=71) was detected at the 54 points by ARUs and UAS

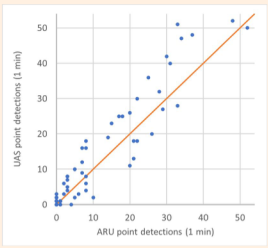


Figure 5. Number of points at which each species (n=71) was detected at the 54 points by ARUs and UAS, comparing only the first minute of ARU data with the UAS data





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Results contd.

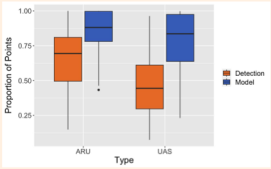


Figure 6. Proportion of points with species detections across the 28 commonest species ("Detected"), and occupancy model estimates of actual proportions occupied ("Model")

- Detection rates were lower for common species on UAS surveys than ARU surveys (Figure 6, orange boxes)
- **BUT,** estimated occupancy for UAS was similar to ARU (Figure 6, blue boxes). *Occupancy Models work!*

Limitations and Future Work

- Sample size → more repetitions beneficial for more accurate occupancy modeling.

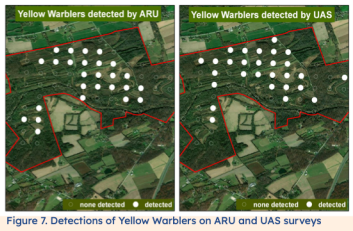


Figure 7. Detections of Yellow Warblers on ARU and UAS surveys

- Some species have very specific habitat requirements, for example the Yellow Warbler *Setophaga petechia* avoids enclosed woodland (Fig. 7).
- We will incorporate habitat covariates into our occupancy models to see whether that improves model fit.

Conclusion

- **1 minute UAS surveys DO provide comparable data to 10 minute ARU surveys!**

Acknowledgements

- Thank you to Gettysburg College, The ES Department, and the X-SIG program for funding our project!

Link to the Summer 2023 X-SIG Research Description:

https://xsigsummer.wordpress.com/2023/06/28/airborne-bioacoustics-is-a-total-hoot-the-use-of-drones-for-bird-conservation/

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